BO CDA 205	Machine Learning Techniques

Course Description:
Supervised learning: decision trees, nearest neighbor classifiers, generative classifiers like naive Bayes, linear discriminate analysis, Support vector Machines, feature selection techniques: wrapper and filter approaches, back-ward selection algorithms, forward selection algorithms, PCA, LDA
Unsupervised learning: K-means, hierarchical, EM, K-medoid, DB-Scan, cluster validity indices, similarity measures, some modern techniques of clustering;
Graphical models: HMM, CRF, MEMM
Semi-supervised learning, Active Learning, Topic Modelling: LDA

Practical component: Lab to be conducted on a 2-hour slot weekly. It will be conducted with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be synchronized with the laboratory classes by using R/Python. The problems to be solved should involve all the algorithm designing techniques that are covered in the theory class.

Learning Resources:

1. Christopher Bishop, Pattern recognition and machine learning, Springer Verlag, 2006.
2. Hastie, Tibshirani, Friedman, The elements of Statistical Learning, Springer, 2009
3. T. Mitchell., Machine Learning, McGraw-Hill, 1997.